single-cell RNA-seq data analysis

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https://github.com/AdrienJolly/TutorialScRNAseq

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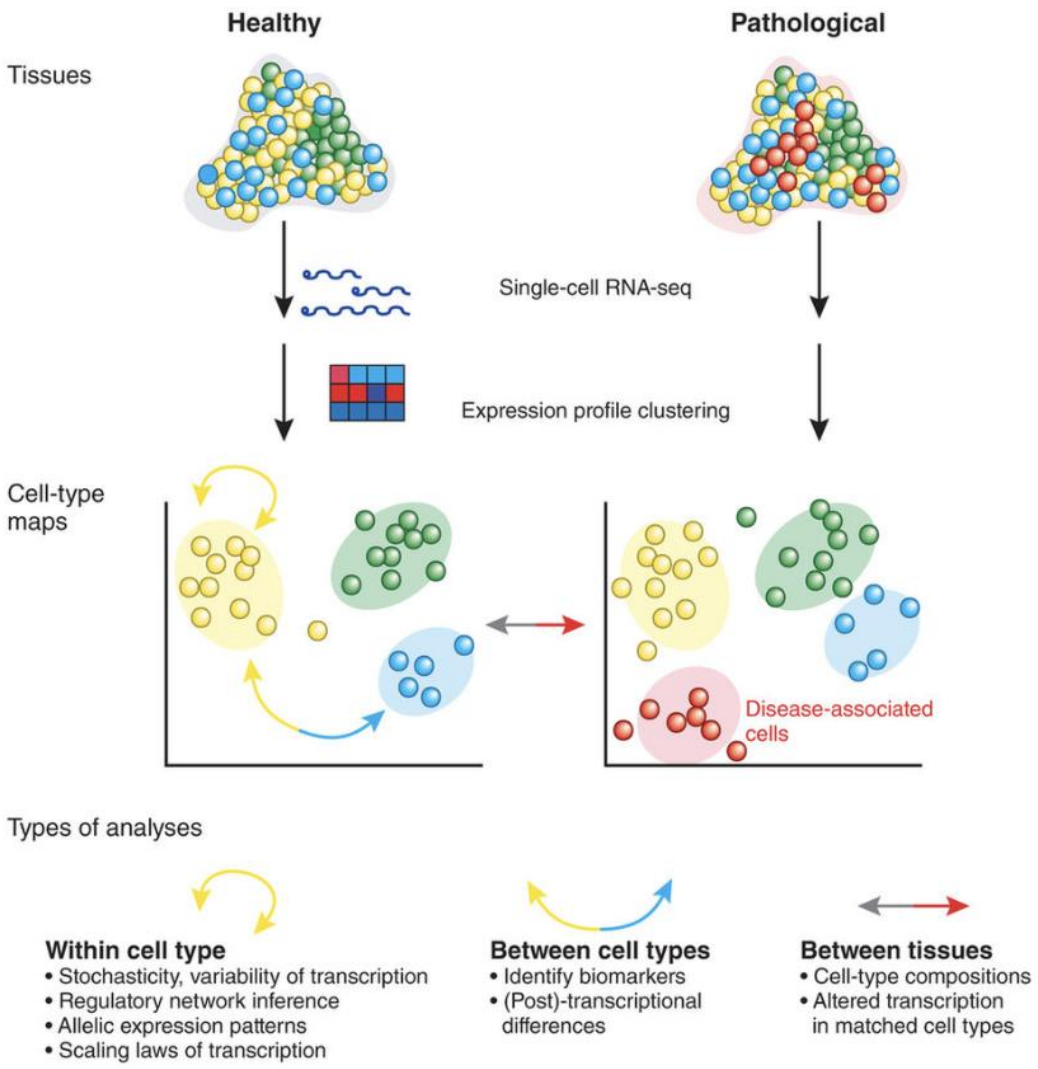
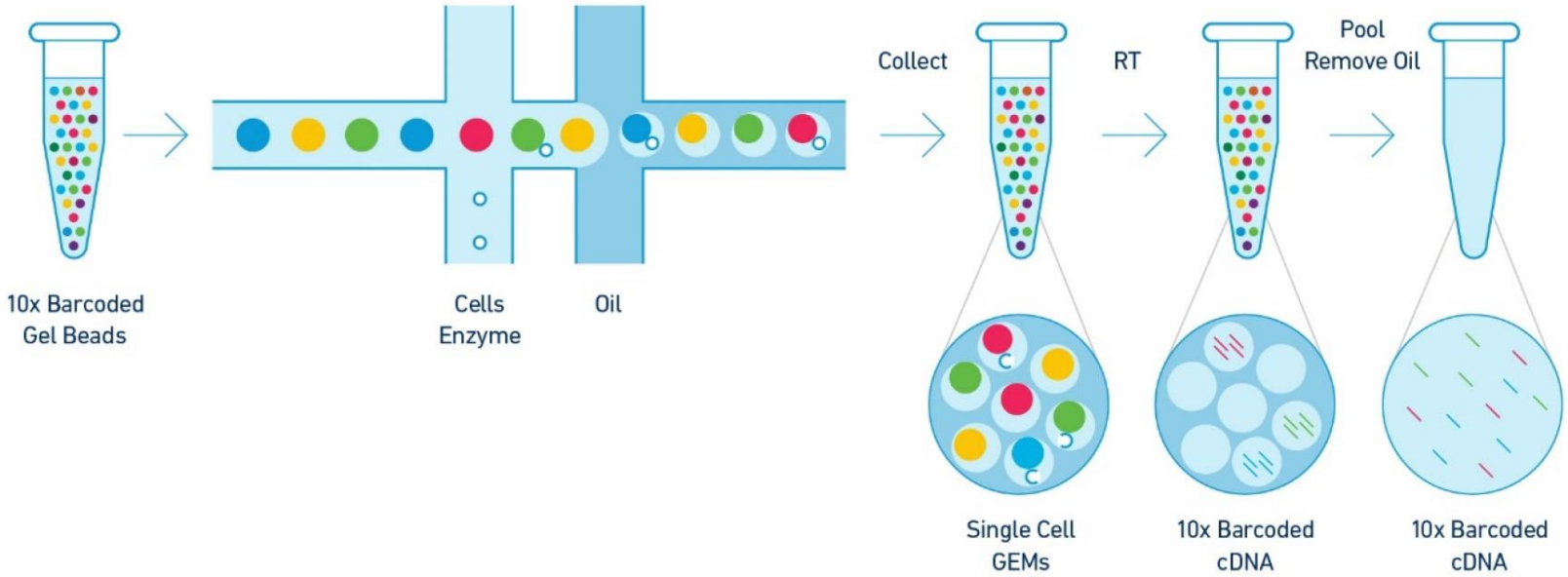
Single-cell RNA-seq: what can we learn from it?



Sandberg. 2014 Nature Methods

10x genomics Chromium technology (droplet based scRNA-seq)



GEM: Gel bead in EMulsion

10x genomics cDNA molecule

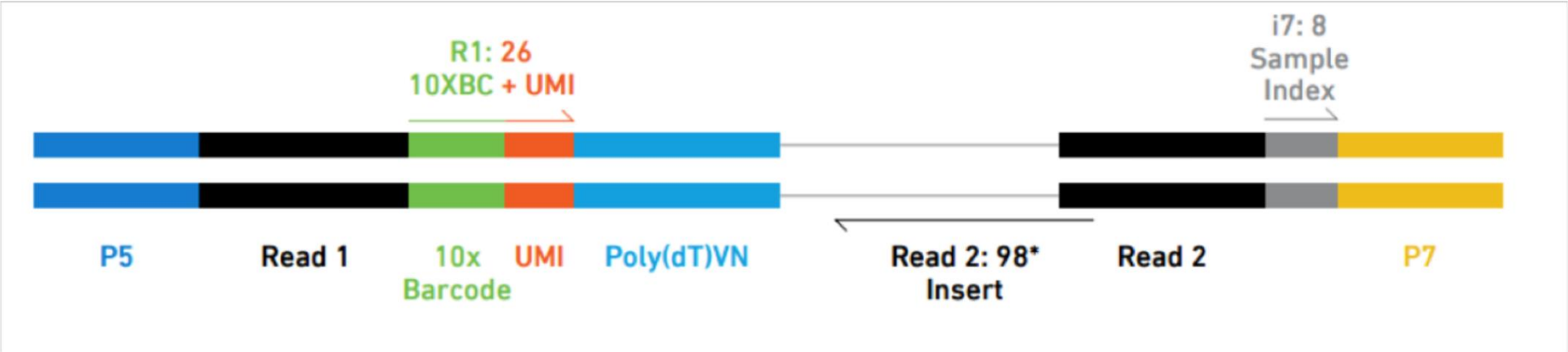


Fig. 2. Schematic of a fragment from a final Chromium™ Single Cell 3' v2 library. *Can be adjusted.

UMI: Unique Molecular Identifier:
The UMI is added before PCR amplification, different for each RNA molecule. Circumvents PCR amplification artifacts

10x Barcode: cell barcode

You only ever get a sample of the cell transcriptome

Limitation:
-RNA capture by chromium
- Sequencing depth
(average number of time a fragment is sequenced)

➢ ~10 000 UMIS per cell vs ~350 000 mRNA in mammalian cell
(https://bionumbers.hms.harvard.edu/)

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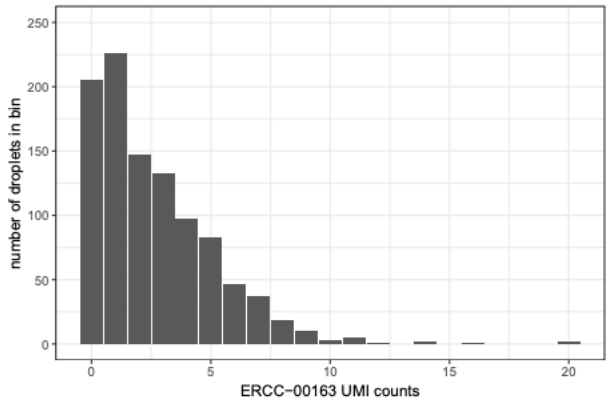
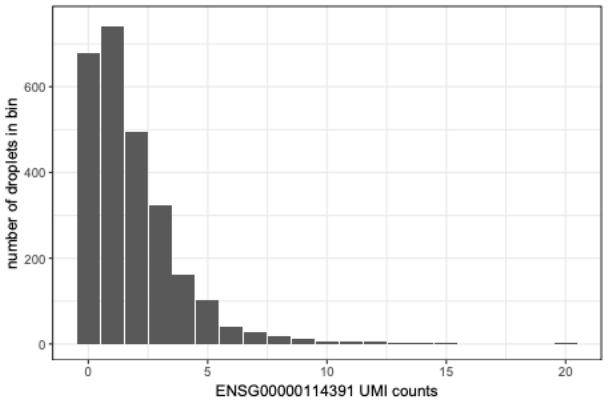
Consequence:
Technical variabiliy
in gene expression level



(c) UMI counts- technical replicates



(d) UMI counts- biological replicates

Towns et al. Genome Biology. 2019

(highly) simplified read preprocessing

FASTQ files (output from sequencing)

Genomic alignment

BAM files

Counting UMI per transcript and cell calling

UMI count matrix Genes*Cells
(What you will start with)

For 10x genomics datasets, it can be performed by CellRanger,
a proprietary tool developed by 10x.

Main toolkits for omics analysis (single-cell, bulk, spatial…)

Bioconductor
OPEN SOURCE SOFTWARE FOR BIOINFORMATICS

https://www.bioconductor.org/

- community of developer for bioinformatics, interoperable packages, high quality documention ()

SEURAT https://satijalab.org/seurat/

- Package developped by Satija lab, integrates all standard functions for single cell analysis

SCVERSE https://scverse.org/

- Python equivalent of Bioconductor, series of packages centered around Scanpy.

Many computationally intensive tools which could not be run on R (factor analysis, deep learning-based).
Almost unavoidable for large datasets (+100 000 cells), where R would be too slow.

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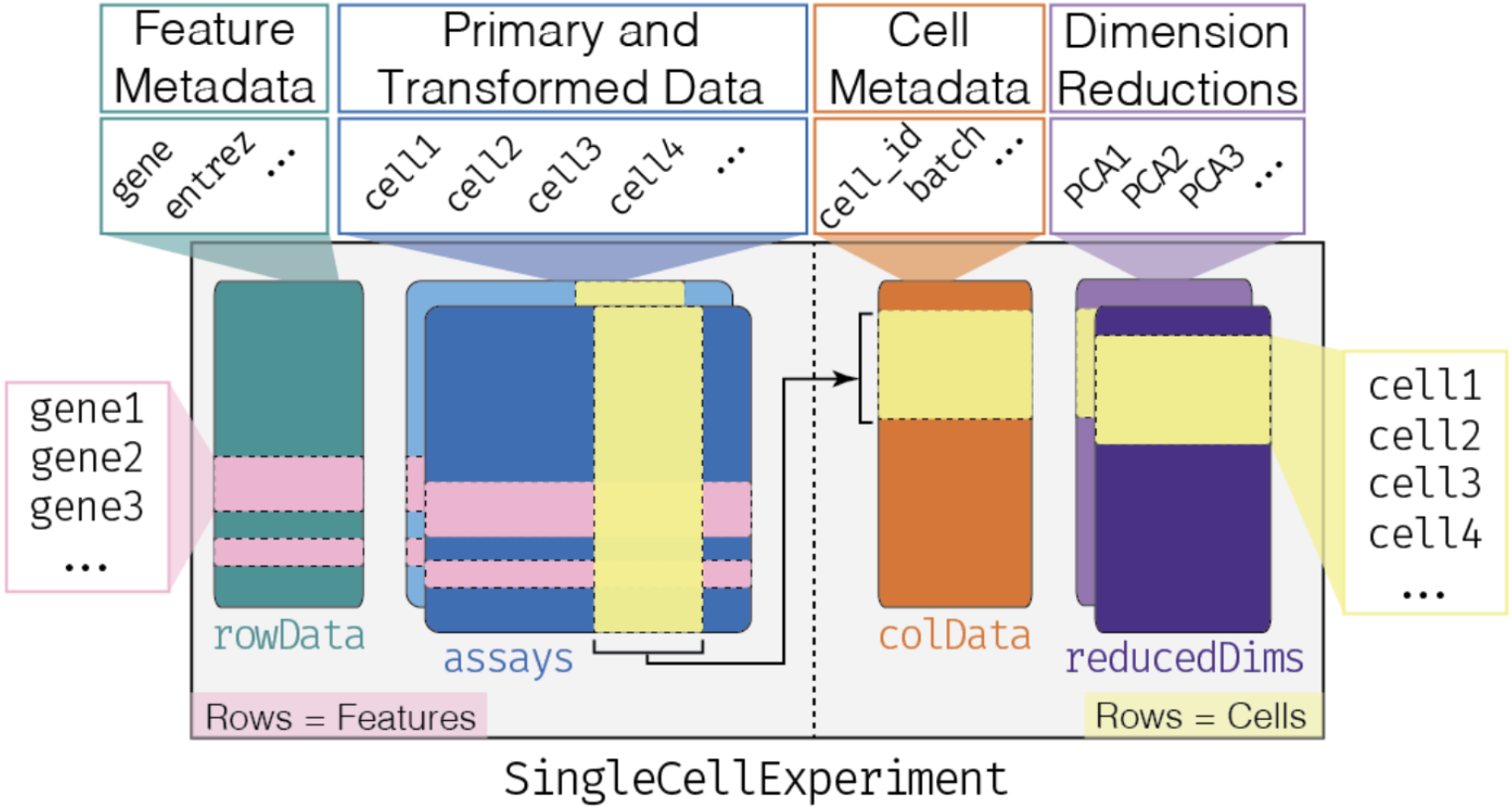
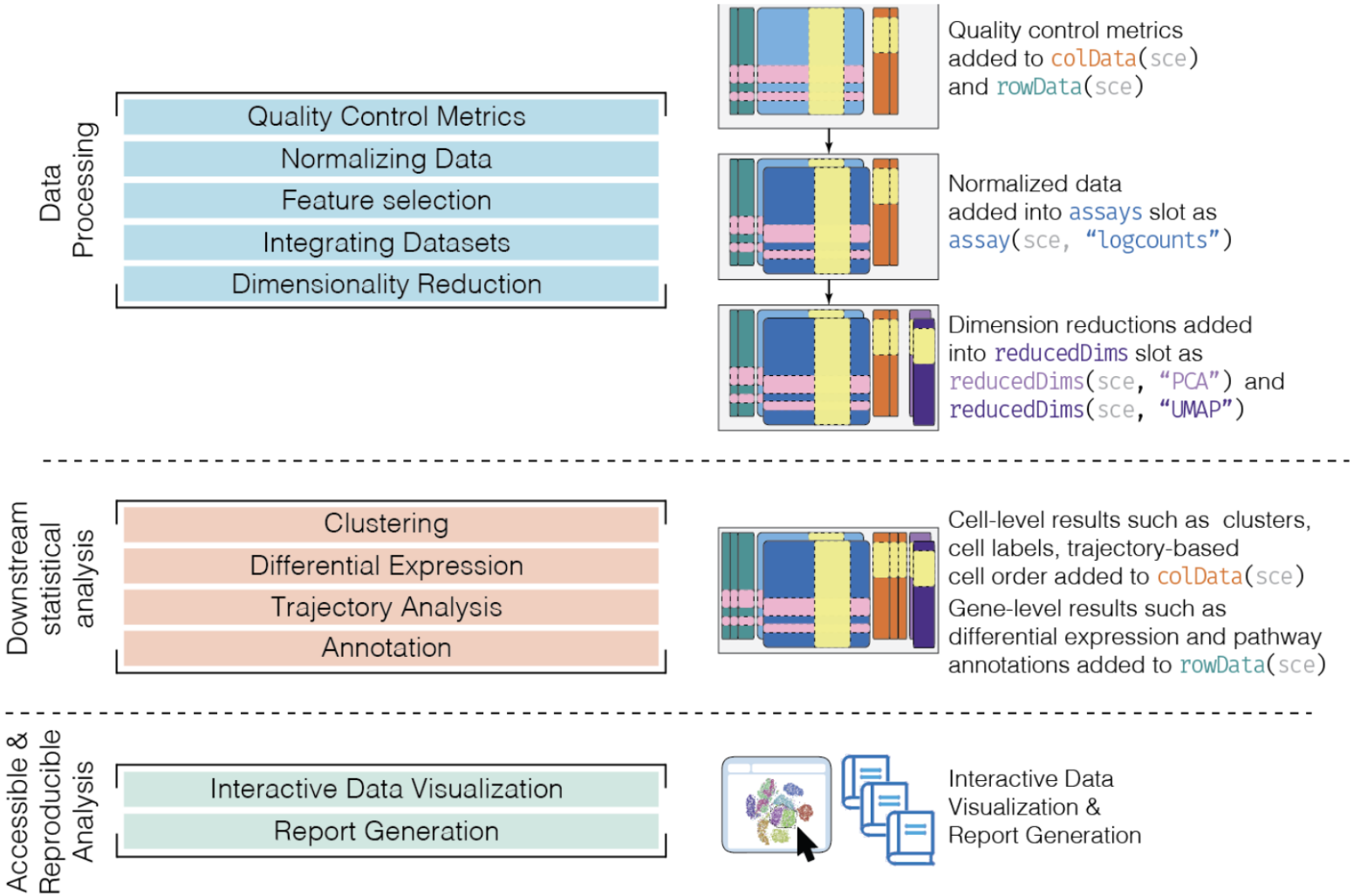
Data infrastructure: The SingleCellExperiment



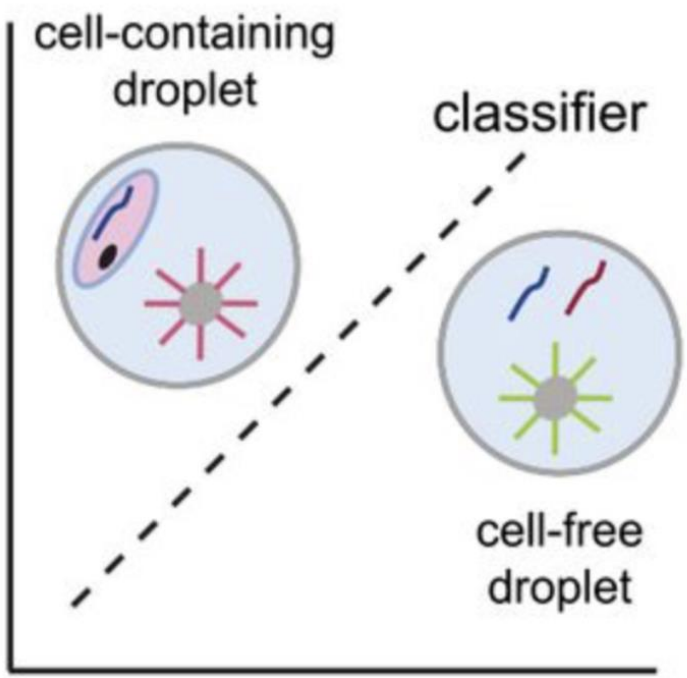
Typical analysis workflow (generally true whatever the bioinformatic tools you use)



Quality control

The goal is to identify and exclude empty **droplets**, **low quality cells** and **doublets**

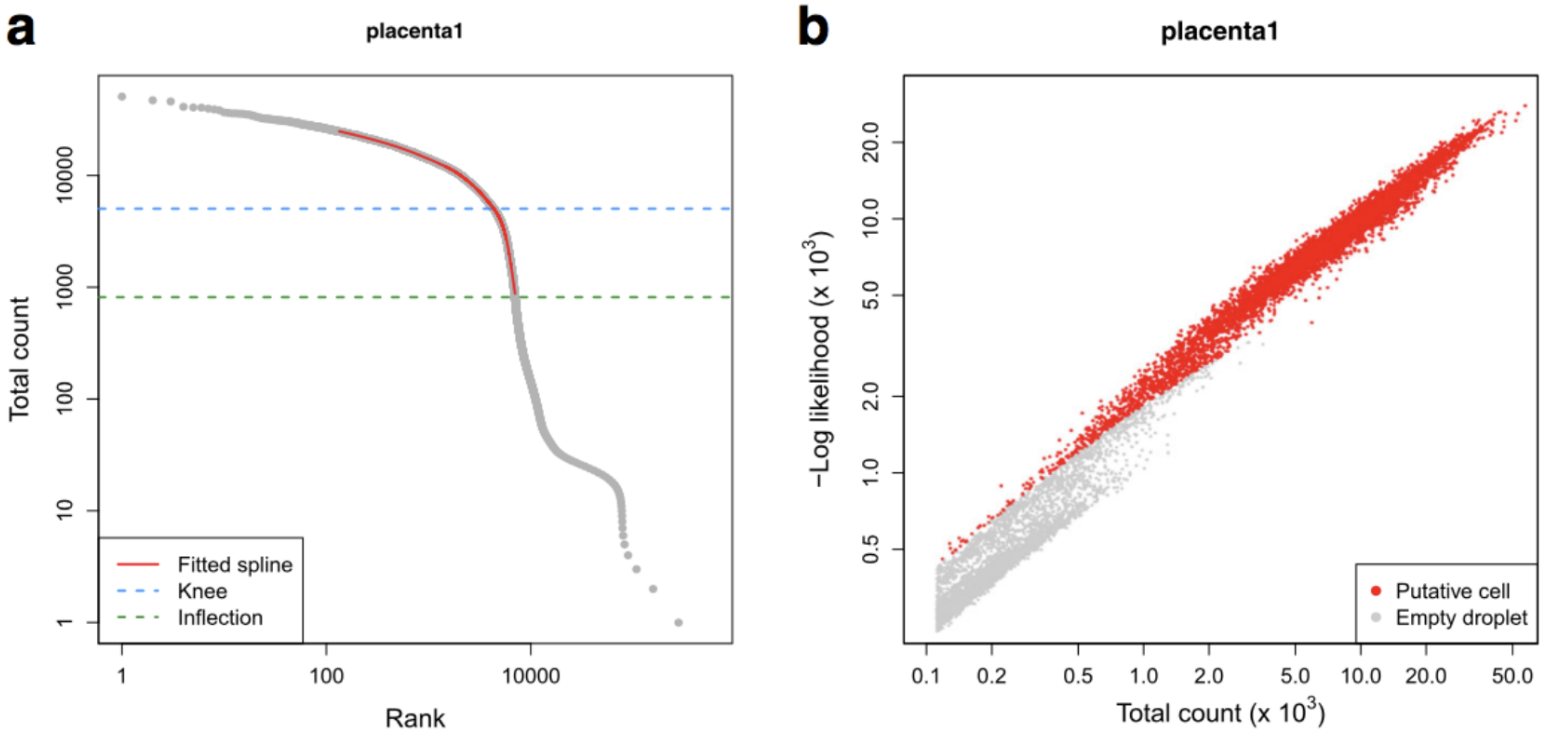
Empty droplets



In droplet-based methods, you sequence both beads
carrying cells and empty droplets
which may have contaminations due to capture of ambient RNA

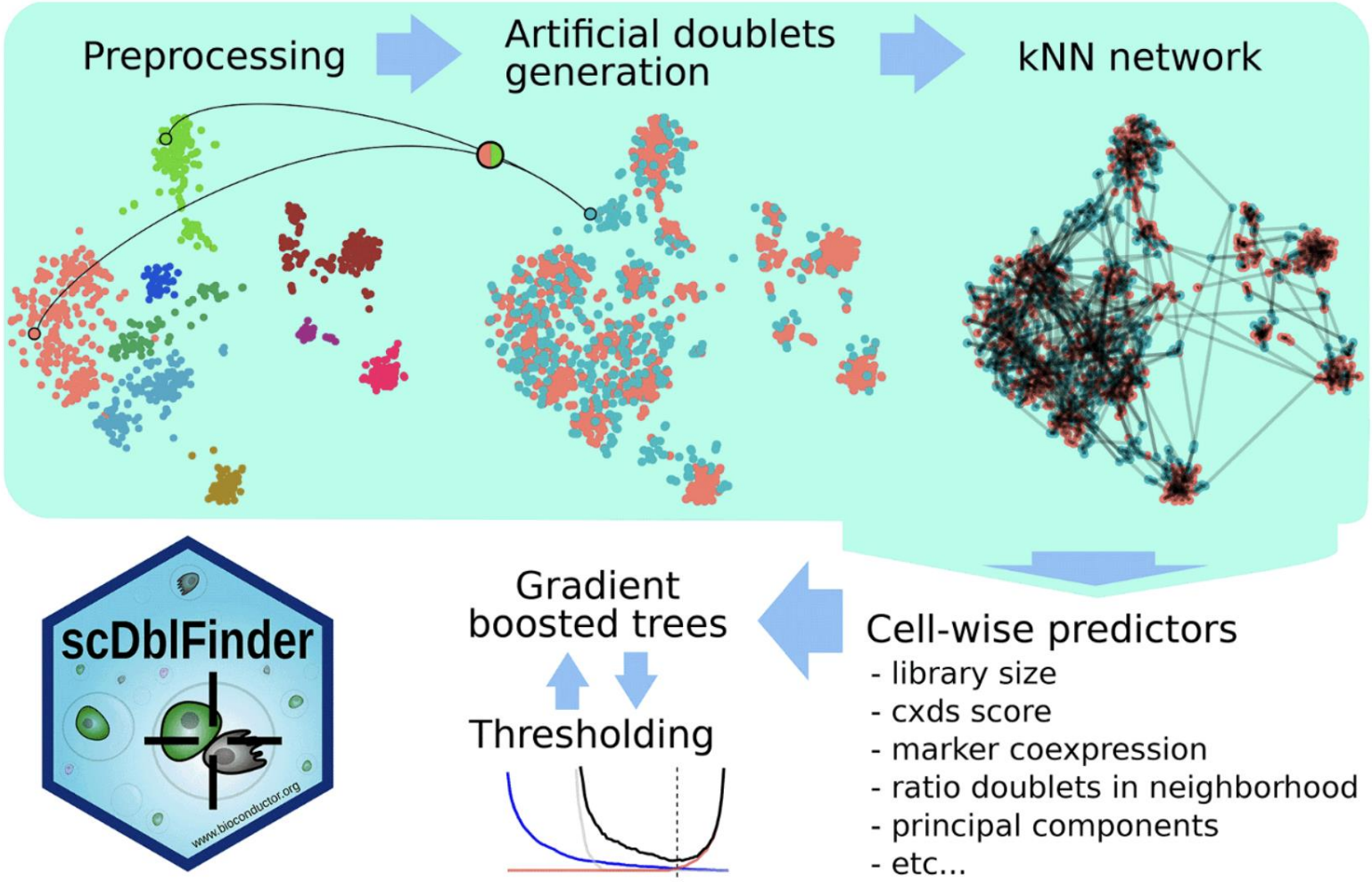
We'll use a statistical method from the package dropletUtils
To:
1/ estimate ambient noise using droplets containing few RNAs
2/ for each droplet, test whether the UMI distribution is consistent with noise exclusively

Empty droplets vizualisation



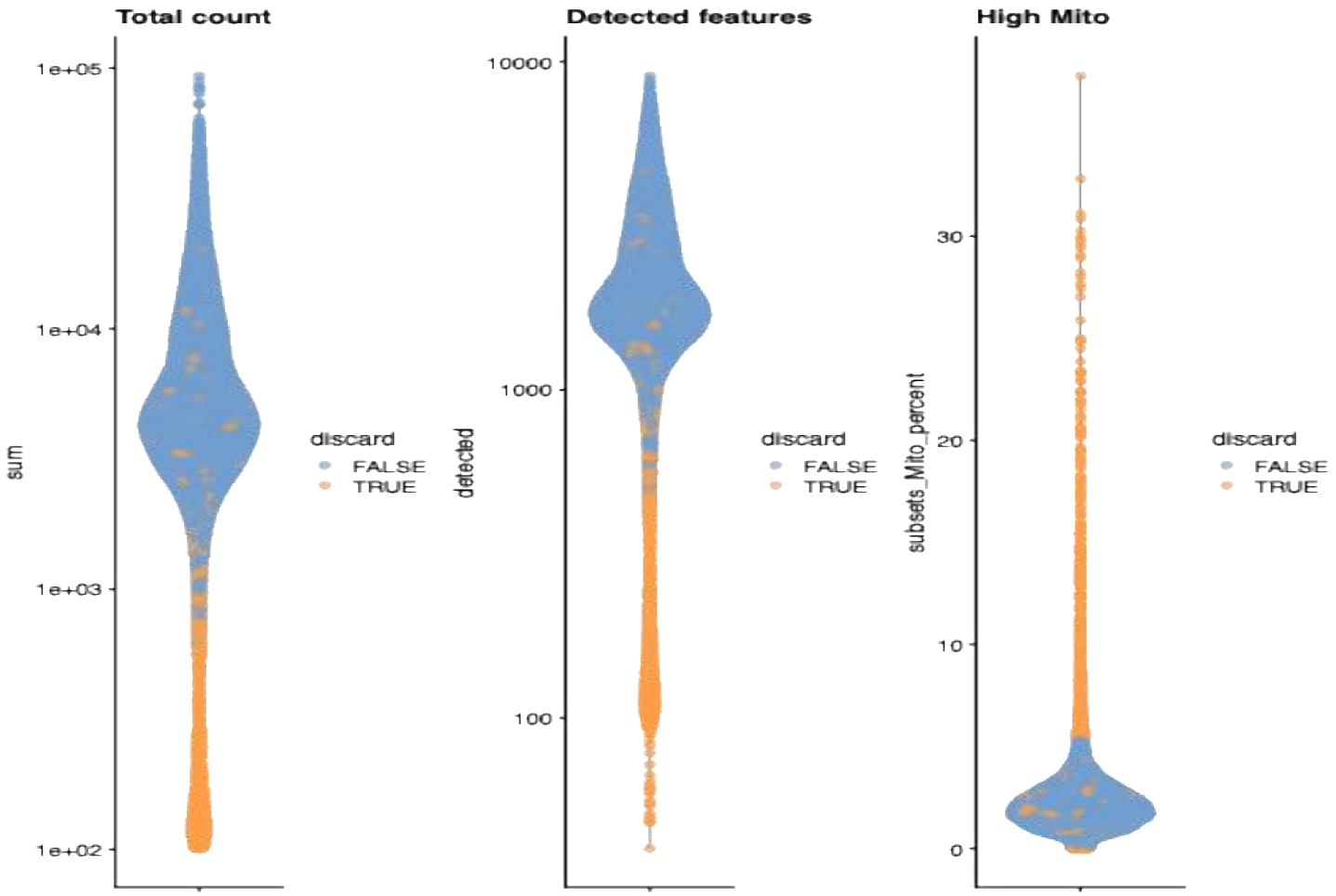
Empty droplets can be visualized
Using elbow/knee plot (a)

Doublet calling



Germain et al. 2021

Low quality cell exclusion



We want to exclude low quality cells with low UMI counts (**total count**), low number of **detected genes** Or **high % of mitochondrial genes**, indicative of dying cells

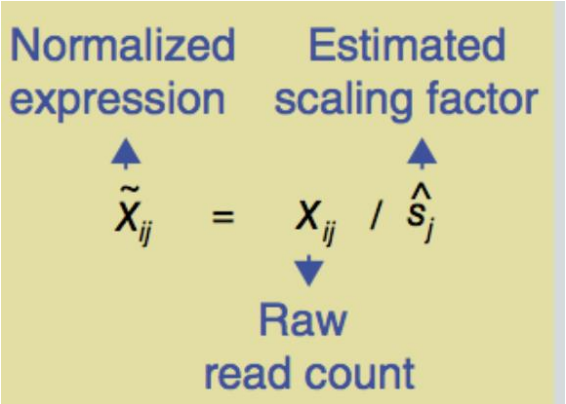
These cells if kept may distort the dataset (clustering), identification of differentially expressed genes

We use a statistical method for outlier detection from the Package scran

Normalization

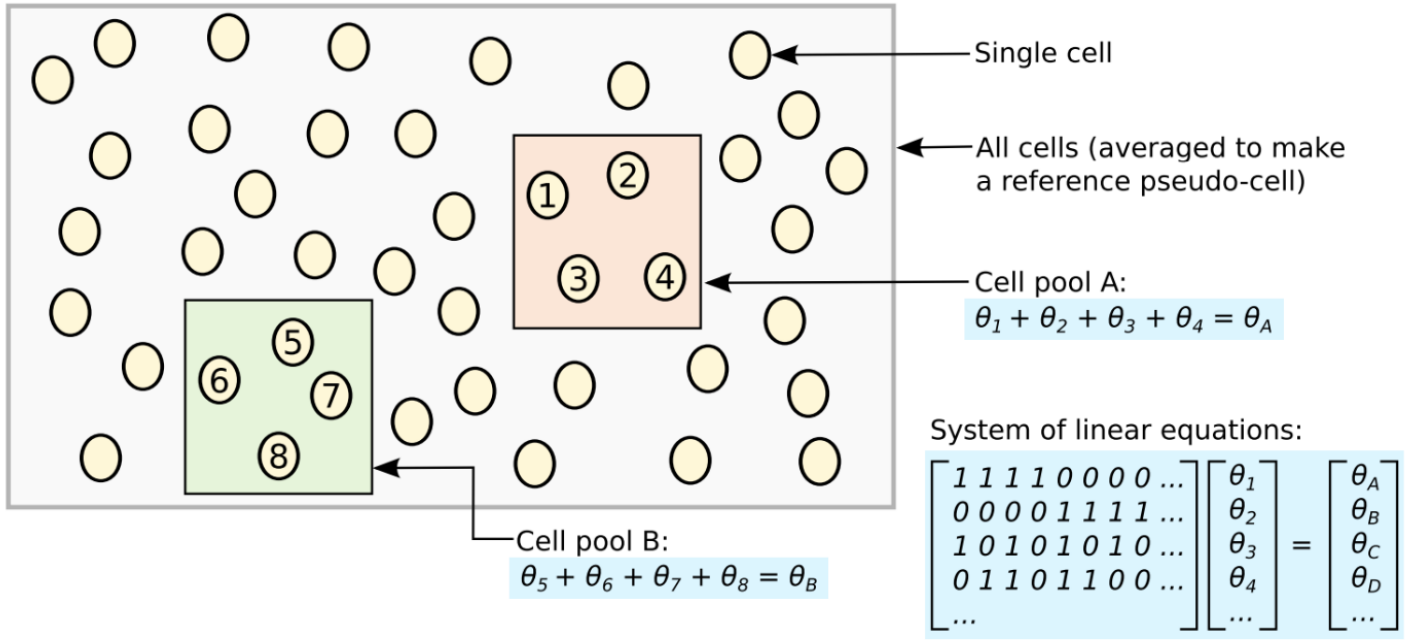
The library size per cell (number of UMI) can be affected by RNA capture, by PCR amplification and by **sequencing depth**

To compare cells between them we must rescale their transcriptomeso that:



Typically, the normalized counts are then Log2 transformed to reduced the correlation between variance and mean

Normalization by deconvolution (scran)



Due to the high number of 0s (non expressed genes) in Sc RNA-seq, one cannot just scale each cell by the average library size, so similar cells are pooled together, a scaling factor is estimated per pool, and the the pooled is deconvolved.

Lun et al 2016. Genome BIology

Dimensionality reduction

Serves 2 distinct purposes:

- infer low dimensional signal from high-dimensional data
- Visualize data

Principal Component Analysis (PCA)

- Linear transformation which reduces the dataset into principal components representing axes of variation (S phase of the cell cycle, IFNa response,…)
- Instead of considering the expression of single genes, you consider the expression of linear combinations of genes (components)
- Each components are orthogonal to each other (uncorrelated)
- Components can be ranked based on how much variance in the data they explain, allowing the user to exclusively consider significant sources of variation

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➢ 20 components or more may provide important information for a scRNA-seq dataset which limits the suitability of PCA for visualization

UMAP (Uniform Manifold Approximation and Projection)

- It builds a neighborhood graph based on transcriptomic distance between cells and projects the cells on a 2D or 3D embedding
- It strives to preserve neibghborhood relationships between cells dut does not conserve true distances.
- Actual distance between cells is affected by an adjustable parameters: ”n neighbors”
- Good to identify communities of cells visually and therfore great for visualisation but not ideal for clustering since distances are not well conserved.

Now let’s start the analysis!